

Symptom: Hostname resolution fails entirely — client can reach a server by IP but not by name

Overview

Basic network connectivity works (ping by IP succeeds), but attempting to resolve a hostname to an address fails outright. This isolates the problem specifically to the DNS resolution step, ruling out routing or firewall issues affecting the actual connection.

Cause 1: Client Not Reaching Its Configured DNS Server

If the DNS server address configured on the client is wrong, unreachable, or was assigned via a now-invalid DHCP lease, every resolution attempt will time out or fail, regardless of whether the DNS server itself is healthy.

Diagnostic steps:

- Confirm the client's configured DNS server IP (ipconfig /all or /etc/resolv.conf) and test direct reachability to it (ping, or a manual query with nslookup/dig specifying that server explicitly)
- If the DNS server address came from DHCP, verify the DHCP scope is still handing out the correct, currently-valid DNS server address

Cause 2: DNS Server Not Forwarding External Queries

An internal DNS server that resolves internal hostnames correctly but has no configured forwarder (or an unreachable one) for external/internet domains will fail specifically on external lookups while internal name resolution continues working fine.

Diagnostic steps:

- Test resolution of both an internal hostname and a known external domain separately to isolate whether the failure is scoped to external queries only
- Check the DNS server's configured forwarders and confirm they're reachable and responding

Symptom: Some domains resolve correctly, others intermittently fail or return inconsistent results

Overview

Unlike total resolution failure, this pattern — some domains fine, others flaky — points toward caching behavior, a specific misbehaving authoritative server, or a DNS server capacity issue rather than a client-wide misconfiguration.

Cause 1: Negative Caching of a Stale Failure

DNS servers cache not just successful answers but also failures (negative caching, per RFC 2308) for a configured TTL. If a domain briefly failed to resolve — even due to a transient upstream issue — the failure itself can get cached and continue being served back to clients well after the underlying problem is resolved.

Diagnostic steps:

- Check the negative caching TTL configured on the resolving DNS server
- Manually flush the DNS cache on the resolving server (and, if applicable, the client) and retest to see if the issue clears immediately, which would confirm stale negative caching as the cause

Cause 2: Authoritative Server for That Specific Domain Is Degraded

If the inconsistency is isolated to one or a few specific domains rather than being general, the problem may not be with your DNS infrastructure at all — it could be an availability or load issue with the authoritative nameserver responsible for that particular domain.

Diagnostic steps:

- Query the domain's authoritative nameservers directly (bypassing your local resolver) to see if the inconsistency is reproducible at the source
- Check whether the affected domain(s) share a common authoritative DNS provider, which would point to a provider-side issue rather than your own infrastructure

Symptom: DNS queries are slow — resolution eventually succeeds, but takes noticeably longer than expected

Overview

Unlike outright failure, this is a latency problem in the resolution path itself, which can create a misleading impression of general "slow network" performance when the underlying cause is isolated entirely to DNS lookup time.

Cause 1: Resolver Falling Through Multiple DNS Servers Sequentially

If a client or resolver is configured with a primary DNS server that's unreachable or slow to respond, and only falls back to a secondary server after a timeout, every query pays that timeout penalty before eventually succeeding via the working server.

Diagnostic steps:

- Measure response time from each configured DNS server individually (dig @<server> <domain> or equivalent) to identify which one is slow or unresponsive
- Reorder or remove a consistently slow/unreachable DNS server from the client's configuration rather than leaving it as a always-timing-out primary

Cause 2: Recursive Resolution Taking Multiple Hops for Uncached Domains

The first time any domain is queried (nothing cached anywhere in the resolution chain), the resolving server must walk the full chain from root servers down to the domain's authoritative server, which is inherently slower than a cached answer. This can look like "DNS is slow" when it's actually normal first-query behavior for infrequently accessed domains.

Diagnostic steps:

- Compare response time for the same query on a first attempt versus a repeated attempt shortly after — if the second attempt is fast, this confirms normal caching behavior rather than an actual problem
- If even repeated queries for common domains remain slow, that points away from this cause and toward server load or path latency instead

Symptom: DNS resolves to an unexpected or outdated IP address

Overview

The lookup itself succeeds — no timeout, no error — but the returned address is wrong: pointing to a decommissioned server, an old IP after a migration, or an address that doesn't match what's expected. This is a data-accuracy problem, not a resolution-failure problem.

Cause 1: Stale Cached Record (TTL Not Yet Expired)

After a DNS record is intentionally updated (e.g., during a server migration), any resolver or client that cached the old record will continue serving it until that record's TTL expires — which can be minutes, hours, or longer depending on how the record was originally configured.

Diagnostic steps:

- Check the TTL that was configured on the record before the change — a long TTL (e.g., 24 hours) set before a planned migration is a common, avoidable cause of this exact symptom
- Query the domain's authoritative nameserver directly to confirm it's actually serving the new, correct value (ruling out a server-side misconfiguration) versus the issue being purely propagation delay from caching

Cause 2: Split-Horizon DNS Returning the Wrong Zone's Answer

In environments using split-horizon DNS (different answers served depending on whether the query originates internally or externally), a client that's expected to receive the internal answer but is actually being resolved via the external-facing zone (or vice versa) will get a technically valid but contextually wrong address.

Diagnostic steps:

- Confirm which DNS view/zone actually answered the query, and compare against which one the client should be using based on its network location
- Test the same query explicitly against both the internal and external resolvers to see if they return different results, confirming a split-horizon configuration is in play

Symptom: DNS lookups return a "SERVFAIL" or other explicit error response rather than timing out or resolving

Overview

Unlike a silent timeout, an explicit error response (SERVFAIL, REFUSED, or similar) means a DNS server was reached and responded — it's actively declining or failing to answer, which points to a server-side configuration or health issue rather than a connectivity problem.

Cause 1: DNSSEC Validation Failure

If DNSSEC validation is enabled on the resolving server and a domain's DNSSEC signatures are broken, expired, or misconfigured (a fairly common occurrence when a domain owner rotates signing keys incorrectly), the resolver will correctly refuse to return a result it can't cryptographically validate — returning SERVFAIL instead.

Diagnostic steps:

- Test the same query with DNSSEC validation temporarily disabled (or use a resolver known not to validate) to see if the domain resolves successfully — if it does, this strongly points to a DNSSEC signing issue on the domain's side, not your infrastructure
- Use a DNSSEC-specific diagnostic query to check the validation chain for the affected domain

Cause 2: Zone Configuration Error on the Authoritative Server

A malformed zone file, a missing required record (like a properly configured SOA or NS record), or a recent botched edit to zone data can cause an authoritative server to return SERVFAIL for queries against that zone, even though the server itself is otherwise healthy and reachable.

Diagnostic steps:

- Check the authoritative server's logs at the time of the failure for a specific zone-loading or parsing error
- Validate the zone file's syntax using the DNS server software's built-in zone-check utility before assuming a network-level cause